

Question:

Which type of articulators permit horizontal as well as vertical motion, but do not orient the motion to the TMJ via facebow transfer?

Answer Choices:

- A. Class I
- B. Class II
- C. Class III
- D. Class IV

Reasoning + Answer:

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To determine which type of articulator allows horizontal as well as vertical motion but does not orient the motion to the TMJ via facebow transfer, we must understand the classification of dental articulators.

Class I articulators:

- Allow both vertical and horizontal movements.
- Motion is oriented based on facebow transfer, which aligns the articulator with the TMJ.
- Therefore, they do orient motion via facebow transfer - does not fit.

Class II articulators:

- Allow both vertical and horizontal movements.
- Motion is oriented based on facebow transfer - does not fit.

Class III articulators:

- Allow both vertical and horizontal movements.
- Do not require facebow transfer; they are designed to allow motion without orientation to the TMJ.
- The motion is not aligned with the TMJ via facebow transfer.
- This matches the question's criteria.

Class IV articulators:

- Typically have limited motion and are not designed for full horizontal and vertical motion.
- Often used for specific functions, not general TMJ simulation.

Thus, the only articulator that allows both horizontal and vertical motion and does not orient the motion via facebow transfer is the Class III articulator.

Class III articulators are known for their ability to provide full-range motion (both vertical and horizontal) without relying on facebow transfer to orient the motion to the TMJ. This makes them suitable for research or when TMJ orientation is not required.

Therefore, the correct answer is Class III.

<answer>

C. Class III

</answer>